Data Structures



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Learning Outcomes

- On completion of this session, you should be able to
  - Describe the term atomic data type
  - Describe the term composite data type
  - Explain the differences between atomic and composite data types
  - Define the term data structure
  - Explain data, structure and operations associated with a data structure

Atomic Data Type

- A set of non-decomposable data
- Consists of
  - A set of homogeneous data
  - A set of operations that can be performed on data

Composite Data Type

- A set of decomposable data
- Composed using atomic and composite data types
- Could consist of a set of
  - Homogeneous or
  - Heterogeneous data

Data Structure

- A structure that holds a set of atomic or composite data
- Consists of
  - A set of homogeneous or heterogeneous data
  - A set of operations that can be performed on data

Questions